# Formalized Class Group Computations and Integral Points on Mordell Elliptic Curves

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## Abstract

Diophantine equations are a popular and active area of research in number theory. In this paper we consider Mordell equations, which are of the form $y^2 = x^3 + d$, where $d$ is a (given) nonzero integer number and all solutions in integers $x$ and $y$ have to be determined. One non-elementary approach for this problem is the resolution via descent and class groups. Along these lines we formalized in Lean 3 the resolution of Mordell equations for several instances of $d < 0$. In order to achieve this, we needed to formalize several other theories from number theory that are interesting on their own as well, such as ideal norms, quadratic fields and rings, and explicit computations of the class number. Moreover, we introduced new computational tactics in order to carry out efficiently computations in quadratic rings and beyond.

***Keywords:*** formalized mathematics, algebraic number theory, Diophantine equations, tactics, Lean, Mathlib

## 1 Introduction

The study of Diophantine equations forms a large part of modern algebraic number theory and arithmetic geometry [? ]. Compared to related areas of mathematics, relatively little of this topic has been covered in an interactive theorem prover. This could be in part due to the lack of formalized prerequisites necessary for work beyond elementary number theory, such as the theory of algebraic number fields including class and unit groups. Additionally, many Diophantine equations require explicit calculations for their effective resolution. Such calculations are nowadays performed by computer algebra systems that have not been formally verified, like Mathematica, Magma, Pari/GP, and SageMath.

In this paper, we study the family of Diophantine equations known as Mordell equations,

$$y^2 = x^3 + d, \quad x, y \in \mathbf{Z}, \tag{1}$$

and formalize a full description of the solutions for various instances of the nonzero parameter $d \in \mathbf{Z}$ (Section ??). We classify the solutions, building on some fundamental algebraic number theory (Section 2), using explicit calculations for quadratic number fields (Section ??) and class groups (Section ??). Such calculations have not been carried out previously in a proof assistant. The complexity of these arguments necessitates careful setup of the theory and developing abstractions to prove these results without extreme overhead.

We report on how the formalization is carried out and highlight three important outputs of our work: First of all, there is the formalization of the explicit solution of some nontrivial Diophantine equations. Secondly, we develop several other theories of independent interest, such as a theory of quadratic fields, ideal norms, and class group computations. Thirdly, we also set up computational structures and add tactics for performing calculations in extensions of commutative rings (Section ??) and interacting with computer algebra systems (Section ??). Wherever possible, we aimed to work at a level of generality that applies beyond that needed for applications to the Mordell equations we considered. Noteworthy difficulties that our formalization resolved are developing general theory and calculating with specific numbers in the same setting of quadratic rings, side conditions being resolved through computations in the typeclass system (Sections ?? and ??), and avoiding more advanced geometry of numbers through an explicit method of calculating the class group (Section ??). We hope that this will pave the way for further formalization developments in (computational) algebraic number theory, and the explicit resolution of many other interesting Diophantine equations in particular.

The basic theory of Dedekind domains up to the finiteness of the class group of global fields was previously formalized [? ] as part of the `mathlib` library [? ] (using version 3 of the Lean theorem prover [? ]). We build on that work here, explicitly computing class groups of several quadratic number

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fields as part of our effort, and fleshing out the surrounding theory. Full source code of our formalization is available online.[1]

## 2 Mathematical background

One of the basic tools we can use to investigate the solutions to a Diophantine equation, especially to show it has no solution, is the method of descent. The main idea is to assume that a solution exists, and choose it to be minimal in some sense. Then, transform this solution in order to 'descend' to another solution to the equation that is smaller than the previous one, contradicting its minimality. If the transformation can be unconditionally applied, the equation has no solutions; otherwise, if the equation has solutions, it will lead to conditions on the solutions.

This method was used by Fermat to prove one instance of his famous Last Theorem, namely that the equation

$$x^4 + y^4 = z^4 \qquad (2)$$

has no nonzero integer solutions. More precisely, Fermat proved ([**?** , Section 1.6]), in his notes on Diophantus' *Arithmetica*, that there exists no right triangle with square area. This can be restated as the non-existence of a nonzero integer solution to the equation $a^4 - b^4 = c^2$, from which in turn the non-existence of a nonzero solution to (2) follows.

The key step in the descent method is the application of the following property: if $a, b, c$ are three integer numbers and $n$ is a natural number such that $\gcd(a, b) = 1$ and $ab = c^n$, then there exist $c_1, c_2 \in \mathbf{Z}$ such that $a = \pm c_1^n$ and $b = \pm c_2^n$. However, it is often convenient to use descent not over the ring $\mathbf{Z}$, but over some larger ring, e.g. the ring of integers of a quadratic number field.

We now recall some useful definitions. We assume some familiarity with basic ring and field theory, as described in many (undergraduate) texts, including [**? ?**]. A *number field* $K$ is a finite extension of the field $\mathbf{Q}$. It is a finite dimensional vector space over $\mathbf{Q}$, and its dimension is called the *degree* of $K$ (over $\mathbf{Q}$). Examples of number fields are $\mathbf{Q}$ itself (of degree 1), $\mathbf{Q}(\sqrt{5}) = \{a + b\sqrt{5} : a, b \in \mathbf{Q}\}$ and $\mathbf{Q}(\sqrt{-5}) = \{a + b\sqrt{-5} : a, b \in \mathbf{Q}\}$ (both of degree 2), and $\mathbf{Q}(\sqrt[4]{17}) = \{a + b\sqrt[4]{17} + c\sqrt[4]{17}^2 + d\sqrt[4]{17}^3 : a, b, c, d \in \mathbf{Q}\}$ (of degree 4). We will mostly use *quadratic fields*, i.e. number fields of degree 2 (Section **??**), as in the second and third example above. In general, any quadratic number field is isomorphic to a field of the form $\mathbf{Q}(\sqrt{d}) = \{a + b\sqrt{d} : a, b \in \mathbf{Q}\}$ for some integer $d \neq 1$ which is squarefree (i.e. not divisible by $p^2$ for a prime $p$). Said differently, these quadratic number fields are constructed by adjoining a (complex) root $\alpha = \sqrt{d}$ of the polynomial $X^2 - d$, which is irreducible over $\mathbf{Q}$ by the conditions imposed on $d$. This generalizes to arbitrary degree $n \in \mathbf{Z}_{>0}$ as follows. For any polynomial of degree $n$ that is irreducible over $\mathbf{Q}$, adjoining any one of its (complex) roots $\alpha$ to $\mathbf{Q}$ will yield the degree $n$ number field $\mathbf{Q}(\alpha) = \{a_0 + a_1\alpha + \ldots + a_{n-1}\alpha^{n-1} : a_0, a_1, \ldots, a_{n-1} \in \mathbf{Q}\}$ (the $n$ different choices of $\alpha$ will all yield isomorphic fields). Conversely, any number field of degree $n$ will be isomorphic to such a field $\mathbf{Q}(\alpha)$.

From an algebraic and arithmetic perspective, the (rational) integers $\mathbf{Z}$ constitute a particularly 'nice' subring of its field of fractions, the rational numbers $\mathbf{Q}$. Upon generalizing from $\mathbf{Q}$ to an arbitrary number field $K$, the analogue of $\mathbf{Z}$ is the *ring of integers* of a number field $K$, denoted $O_K$. It is defined as the integral closure of $\mathbf{Z}$ in $K$:

$$O_K\{x \in K : \exists f \in \mathbf{Z}[X] \text{ monic such that } f(x) = 0\},$$

where we recall that a (univariate) polynomial is called *monic* if its leading coefficient is equal to 1. The fact that $O_K$ is indeed a ring follows for instance from general properties of integral closures ([**?** , Section I.2]).

Some examples of rings of integers are as follows. The ring of integers of $\mathbf{Q}$ is indeed $\mathbf{Z}$. For $K = \mathbf{Q}(\sqrt{d})$, values such as $d \in \{-5, -2, -1, 2, 3\}$ simply give $O_K = \mathbf{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbf{Z}\}$, but certain other values of $d$, e.g. $d \in \{-3, 5\}$, give $O_K = \mathbf{Z}[\frac{1}{2}(1+\sqrt{d})] = \{a+b\frac{1}{2}(1+\sqrt{d}) : a, b \in \mathbf{Z}\}$. That, for example, the ring of integers of $\mathbf{Q}(\sqrt{-3})$ is strictly larger than $\mathbf{Z}[\sqrt{-3}]$ is apparent from the fact that $\frac{1}{2}(1+\sqrt{-3})$ (which is not an element of $\mathbf{Z}[\sqrt{-3}]$) is a root of the monic polynomial with integer coefficients $X^2 - X + 1$. Finally, for $K = \mathbf{Q}(\sqrt[4]{17})$ we have $O_K = \{a + b\sqrt[4]{17} + c\frac{1}{2}(\sqrt[4]{17}^2 - 1) + d\frac{1}{4}(\sqrt[4]{17}^3 + \sqrt[4]{17}^2 + \sqrt[4]{17} + 1) : a, b, c, d \in \mathbf{Z}\}$, which cannot be written in the form $\mathbf{Z}[\alpha]$ for any $\alpha \in O_K$, illustrating the fact that determining rings of integers can become involved rather soon.

The 'nice' properties alluded to earlier, can be made precise by the statement that the ring of integers $O_K$ of a number field $K$ is a so-called Dedekind domain [**?** , Section 2]. While $O_K$ is not necessarily a unique factorization domain (UFD), nonzero ideals in $O_K$ (or more generally in any Dedekind domain) factor uniquely, up to the order of the factors, in (necessarily nonzero) prime ideals.

The key implication in the descent method still holds in general when $D$ is a UFD (replacing $\pm 1$ by a unit in $D$), for example when $D = \mathbf{Z}[i]$ is the Gaussian integers, but is no longer applicable when $D$ is not a UFD, for example when $D = \mathbf{Z}[\sqrt{-5}]$. However, when $D$ is a Dedekind domain such as the ring of integers of a number field, we have a descent for ideals $I, J, L$ in $D$; namely the following implication holds for any natural number $n$:

$$\gcd(I, J) = 1 \wedge IJ = L^n \implies I = L_1^n \wedge J = L_2^n \qquad (3)$$

for certain ideals $L_1, L_2$ in $D$.

To return from ideals back to elements, the *class group* of $D$, denoted $\mathrm{Cl}(D)$ is of crucial importance. Recall that a fractional ideal $I$ of $D$ is a $D$-submodule of $K$ (the field of fractions of $D$) of the form $I = xJ$ with $x \in K$ and $J$ an ideal of

[1] https://github.com/lean-forward/class-group-and-mordell-equation